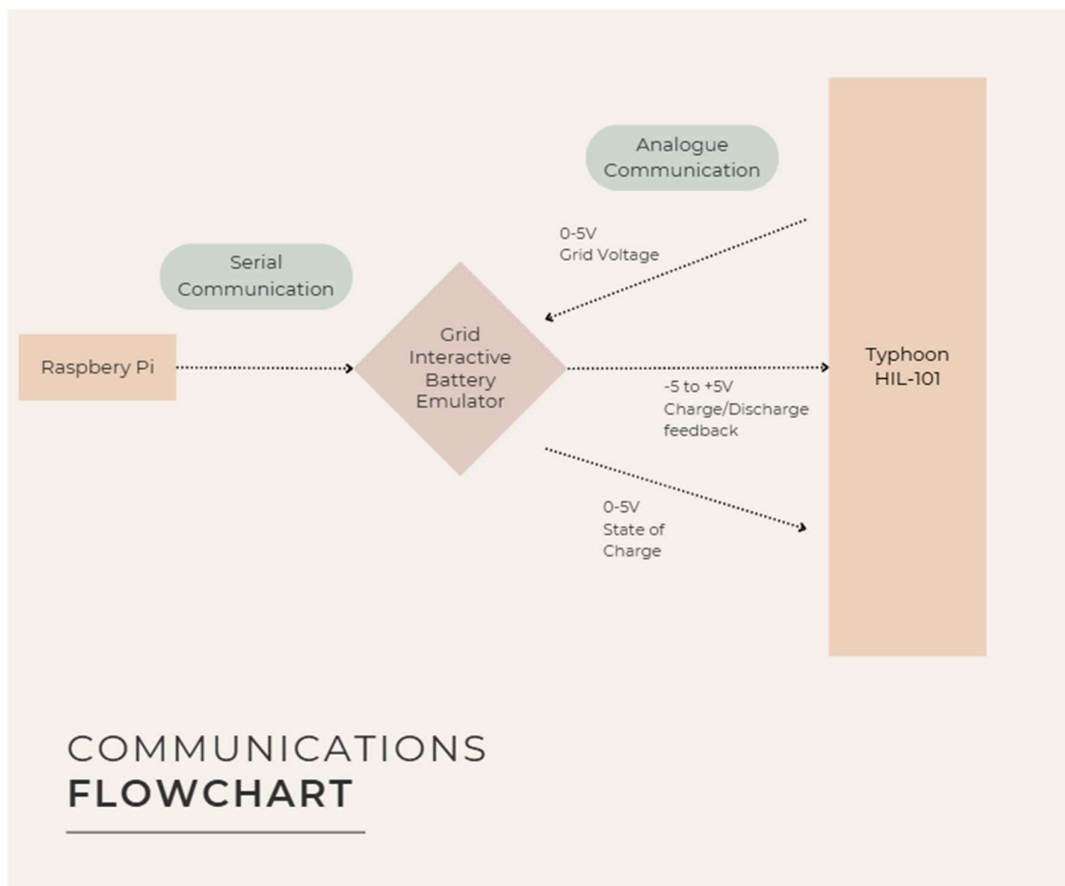


Grid Interactive Battery Emulator

User Manual V1.0



This Battery Emulator is designed to mimic the operation of a typical household grid tied battery. It receives ASCII serial commands (typically from a Raspberry Pi) to either command charging FROM the virtual grid or discharge TO the virtual grid. The analogue outputs on the rear are designed to connect to a Typhoon HIL 101 or similar hardware. These outputs report the battery state and the instantaneous charge/discharge rate. An analogue input also monitors the grid voltage.



Before charging or discharging occurs, the Battery Emulator checks a number of conditions. Should the conditions not be met, the Battery Emulator will not charge or discharge as commanded. For example, the battery cannot charge if the battery is already full or if the grid is too low, it cannot discharge if the battery is empty or if the grid voltage is too high.

Serial protocol:

Serial settings are standard 9600 8N1. That is, 9600 Baud, 8 bits, No parity, One stop bit.

Battery State of Charge (SoC) is reported once per second via the Serial I/O connector. The SoC is reported without any request by the host computer. The output is a 16 bit ASCII string from 00000 to 65500.

The host computer (Raspberry Pi) can command the Battery Emulator to charge or discharge by sending an ASCII string in the form Cxxx or Dxxx where C=Charge, D=Discharge and xxx is a value from 000 to 255.

Rear Panel Connections:



Batt SOC on Startup:

This rotary knob sets the virtual battery SoC soon as power is applied, or just after a power reset. This allows rapid testing of limit states of the host computer software, without having to wait hours for the virtual battery to fully charge or fully discharge.

Auto Chg/Dischg:

-5 to +5V Output. A negative voltage indicates instantaneous Discharging to the grid, while a positive voltage indicates Charging FROM the grid.

Auto Batt V:

0 to 5V Output. Indicates instantaneous SoC of the battery.

Grid 0-5V In:

0 to 5V input (from HIL) indicating grid availability. A nominal 2.5V will indicate a grid that will allow charging and discharging as commanded. A high grid voltage (~4V) will not allow discharging and a low grid voltage (~1V) or less will not allow charging of the battery. No connection to this terminal will allow the internal weak resistor divider to allow for a "Grid nominal" value, in order to stop nuisance limits while developing host computer software.

Man. Chg/Dischg:

-5 to 5V Output, controlled by the “Charge/Discharge Current” knob on the front panel. This feature allows the user to manually send an analogue voltage to the HIL101. It aids in rapid testing of the HIL101 code without yet having written the Raspberry Pi serial communication software.

Man. Batt V.:

0 to 5V Output, controlled by the “Battery Voltage” knob on the front panel. This feature allows the user to manually send an analogue voltage to the HIL101. As above, it aids in rapid testing of the HIL101 code before the Raspberry Pi code is written.